

CH1

Instantiate: 實例化

Calling method

Function could be traditional (not belong to any class) or in a class

Type of methods

accessor: do not change the value

mutator: update the variable instantly

Immutable: 不可變的

The default value instances:

int()	bool()	float()
0	False	0.0

Parameters in Function

formal parameter: identifiers used to describe the expected param

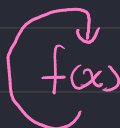
```
def foo(var1, var2):  
    print(var1, var2)  
    return
```

actual parameters: objects sent by the caller when invoking the function

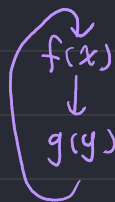
```
foo("Hello", "World")
```

Recursion

direct call



indirect call



- **Definition**

An **algorithm** is a **finite set of instructions** that, if followed, accomplishes a particular task. All the algorithms must satisfy the following criteria:

1. **Input**
 2. **Output**
 3. **Definiteness**
 4. **Effectiveness**
 5. **Finiteness**
- **Definiteness:** each instruction is clear and unambiguous
 - **Effectiveness:** each instruction is executable; in other words, feasibility
可行性
 - **Finiteness:** the algorithm terminates after a finite number of steps.